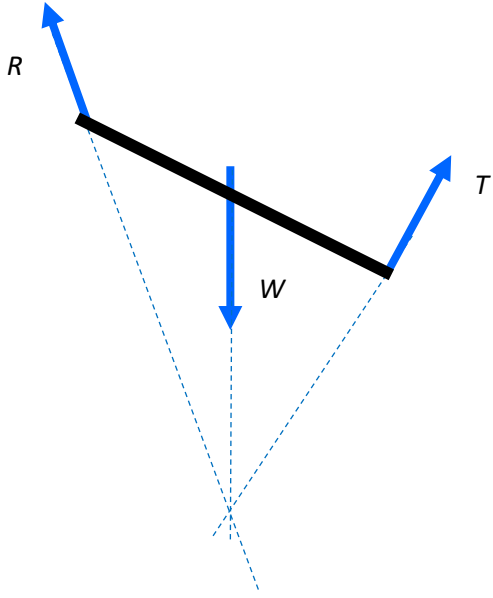
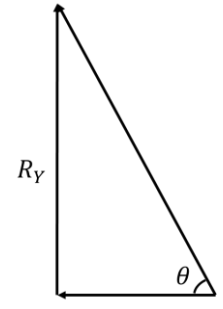


Qn	Answers	Marks
1 (a)	Resultant/ net force is zero Resultant/ net moment about any axis/point is zero	B1 B1
(b)(i)	 • T and W correctly labelled with full name, T and W in correct direction. • All 3 forces intersect to meet at a common point (clearly shown with dotted lines!) and R correctly labelled & in correct direction W is the weight of the rod, T is the tension acting on the rod, R is the force acting on the rod by the hinge/ contact force	M1 A1
(b)(ii)	Take moments about the hinge, $30 \times 9.81 \times \cos 30^\circ \left(\frac{L}{2}\right) = T \times L$ $T = 127.4 \approx 127 \text{ N}$	B2
(b)(iii)	$(\leftarrow+) \quad R_x = T \sin 30^\circ = 127.4 \sin 30^\circ = 63.7 \text{ N}$ $(\uparrow+) \quad R_y = 30 \times 9.81 - T \cos 30^\circ = 184 \text{ N}$ $R = \sqrt{R_x^2 + R_y^2} = \sqrt{63.7^2 + 184^2} = 194.7 \approx 195 \text{ N} \quad \text{ecf allow}$  $\tan \theta = \frac{184}{63.7} \Rightarrow \theta = 70.9^\circ \text{ above the horizontal as shown in Fig. 1 ecf allow}$	C1 C1 A1 A1

$2(e)(i)$	$= 2/T$ (where $T = 10/20 \text{ s} = 17.6 \text{ rad s}^{-1}$)	B1
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